Elementary Analysis

Chapter 2, Section 8

A Discussion about Proofs
Exercises 8.1–8.10, with every subpart

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Text and numbering. Kenneth A. Ross, *Elementary Analysis: The Theory of Calculus*, second edition (Springer, 2013), Chapter 2. Exercise numbers follow the copy already linked on Mathjock. These are independently written worked solutions, not the publisher's solutions manual. Each exercise is identified by a short mathematical description.

Reading philosophy. Every argument states its goal, records the assumptions being used, and explains the intermediate algebra or logical inference. Definitions, sign conditions, and both directions of an "if and only if" are made explicit. A cited rule is named at the point where it is used.

Conventions. $\mathbb{N} = \{1, 2, 3, \ldots\}$, $\mathbb{Z} = \{\ldots, -2, -1, 0, 1, 2, \ldots\}$, and $\mathbb{Q} = \{p/q : p, q \in \mathbb{Z},\ q \neq 0\}$. Unless a different domain is stated, sequence indices run through $\mathbb{N}$ and all terms and finite limits are real. The empty sum is 0, and $0! = 1$.

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8 A discussion about proofs

Every limit proof in this section starts with an arbitrary $\varepsilon > 0$, chooses an appropriate threshold, and verifies the required inequality for every integer index beyond that threshold. An exploratory rearrangement helps discover N; the formal argument then runs forward from $n > N$. Divergence proofs instead show that the definition cannot hold for any proposed real limit. No limit law from Section 9 is assumed.

A proof template for the first exercises

Definition 7.1 says that $x_n \to L$ means

for every real $\varepsilon > 0$ there exists a number N such that every integer $n > N$ satisfies $|x_n - L| < \varepsilon$.

The order of these choices matters: the reader supplies an arbitrary positive ε; we choose N in terms of that ε; then we verify the estimate for *every* sequence index $n > N$. Our choice of N cannot depend on which later index is tested.

Ross permits a real threshold N. If an integer threshold is preferred, the Archimedean property supplies a natural number larger than our chosen N, and the same proof applies to all indices above that larger threshold. The first finitely many terms need not obey the error estimate: the definition requires it only after the threshold.

All the limit assertions below are proved from this definition. Algebraic simplifications, sign rules, and absolute-value identities are used as rules about individual real numbers, without invoking the limit theorems of Section 9.

8.1 Exercise 8.1: Four limits proved directly from the definition

What we must show. Prove each of the four stated limits with a complete ε–N argument.

Part (a): the alternating reciprocal sequence.

We must prove

$$\lim_{n\to\infty} \frac{(-1)^n}{n} = 0.$$

For an integer n, $(-1)^n$ is either 1 or -1, according to the parity of n. Its absolute value is therefore always 1. Since $n \in \mathbb{N}$ implies $n > 0$, the exact error is

$$\begin{aligned}\left|\frac{(-1)^n}{n} - 0\right| &= \frac{|(-1)^n|}{n}, \\ &= \frac{1}{n}.\end{aligned}$$

We need this error to be strictly less than ε.

Let $\varepsilon > 0$ be arbitrary and choose

$$N = \max\left\{1, \frac{1}{\varepsilon}\right\}.$$

Take any integer $n > N$. Then $n > 1/\varepsilon$. Multiplication by the positive number ε gives $\varepsilon n > 1$; division by the positive number n gives $\varepsilon > 1/n$. Consequently

$$\left|\frac{(-1)^n}{n} - 0\right| = \frac{1}{n} < \varepsilon.$$

The estimate holds for every integer $n > N$, as the definition requires.

Conclusion. For every $\varepsilon > 0$ the displayed threshold works, so $\boxed{(-1)^n/n \longrightarrow 0}$. Alternation does not interfere with the estimate because both possible signs have absolute value 1.

Part (b), Step 1: compute the error and the needed threshold.

We must prove $1/n^{1/3} \to 0$, where $n^{1/3}$ is the positive real cube root of n. Since $n^{1/3} > 0$,

$$\left|\frac{1}{n^{1/3}} - 0\right| = \frac{1}{n^{1/3}}.$$

To make this less than ε, it suffices to have $n^{1/3} > 1/\varepsilon$, or $n > 1/\varepsilon^3$. These implications concern ordinary positive real numbers. For example, strict comparison of their cubes follows from

$$u^3 - v^3 = (u - v)(u^2 + uv + v^2) \qquad (u, v > 0).$$

If $u > v > 0$, both factors on the right are positive, so $u^3 > v^3$. Conversely, if $u^3 > v^3$ but $u \le v$, cubing the nonnegative numbers would give $u^3 \le v^3$, a contradiction. Thus comparing positive cubes is equivalent to comparing their positive cube roots.

Part (b), Step 2: give the formal proof.

Let $\varepsilon > 0$ be arbitrary and choose

$$N = \max\left\{1, \frac{1}{\varepsilon^3}\right\}.$$

For an arbitrary integer $n > N$, we have

$$n > \frac{1}{\varepsilon^3} = \left(\frac{1}{\varepsilon}\right)^3.$$

Taking positive cube roots gives $n^{1/3} > 1/\varepsilon$. Multiplying this inequality by $\varepsilon > 0$ and then dividing by $n^{1/3} > 0$ gives

$$\varepsilon n^{1/3} > 1 \quad \Longrightarrow \quad \frac{1}{n^{1/3}} < \varepsilon.$$

Hence the exact error satisfies $\left|1/n^{1/3} - 0\right| < \varepsilon$ for every integer $n > N$.

Conclusion. Definition 7.1 therefore gives $\boxed{1/n^{1/3} \longrightarrow 0}$. The threshold depends on ε^{-3} because the denominator is a cube root.

Part (c), Step 1: simplify the exact difference from the proposed limit.

The sequence in the printed exercise is

$$x_n = \frac{2n - 1}{3n + 2}, \qquad \text{and the proposed limit is } \frac{2}{3}.$$

For every $n \in \mathbb{N}$, $3n+2 > 0$, so the denominator is nonzero. Putting the two fractions over the common denominator $3(3n+2)$ gives

$$\begin{aligned} x_n - \frac{2}{3} &= \frac{3(2n-1) - 2(3n+2)}{3(3n+2)}, \\ &= \frac{6n - 3 - 6n - 4}{9n+6}, \\ &= \frac{-7}{9n+6}. \end{aligned}$$

Because $9n+6 > 0$, taking absolute values yields

$$\left| x_n - \frac{2}{3} \right| = \frac{7}{9n+6}.$$

Also $9n+6 \geq 9n > 0$, so increasing the positive denominator decreases the quotient:

$$\frac{7}{9n+6} \leq \frac{7}{9n}.$$

Part (c), Step 2: choose a threshold and check all later indices.

Let $\varepsilon > 0$ and set

$$N = \max\left\{1, \frac{7}{9\varepsilon}\right\}.$$

If an integer $n > N$, then $n > 7/(9\varepsilon)$. Multiplication by $9\varepsilon > 0$ gives $9\varepsilon n > 7$; division by $9n > 0$ gives $\varepsilon > 7/(9n)$. Thus

$$\left| x_n - \frac{2}{3} \right| = \frac{7}{9n+6} \leq \frac{7}{9n} < \varepsilon.$$

Conclusion. The estimate holds for every integer $n > N$, proving

$$\boxed{\lim_{n\to\infty} \frac{2n-1}{3n+2} = \frac{2}{3}}.$$

Part (d), Step 1: isolate a positive tail of the denominator.

We must prove

$$x_n = \frac{n+6}{n^2-6} \longrightarrow 0.$$

The denominator is nonzero for every positive integer n: if $n \leq 2$, then $n^2 \leq 4 < 6$; if $n \geq 3$, then $n^2 \geq 9 > 6$. For the limit estimate, we will use only indices $n \geq 4$. For such n, we have $n^2/2 \geq 8 \geq 6$, and therefore

$$n^2 - 6 \geq n^2 - \frac{n^2}{2} = \frac{n^2}{2} > 0.$$

This is a lower bound on the denominator, with its sign recorded.

Part (d), Step 2: bound the numerator and then the entire error.

For $n \geq 1$, the inequality $6 \leq 6n$ gives

$$n + 6 \leq n + 6n = 7n.$$

If $n \geq 4$, both the numerator $n+6$ and denominator n^2-6 are positive. Thus

$$\begin{aligned} |x_n - 0| &= \frac{n+6}{n^2-6}, \\ &\leq \frac{7n}{n^2-6} && \text{by the upper bound on the numerator,} \\ &\leq \frac{7n}{n^2/2} && \text{by the positive lower bound on the denominator,} \\ &= \frac{14n}{n^2} = \frac{14}{n}. \end{aligned}$$

The cancellation in the last line is valid because $n > 0$.

Part (d), Step 3: give the formal threshold.

Let $\varepsilon > 0$ and choose

$$N = \max\left\{4, \frac{14}{\varepsilon}\right\}.$$

An arbitrary integer $n > N$ satisfies $n > 4$, so every preceding tail estimate applies. It also satisfies $n > 14/\varepsilon$; multiplying by $\varepsilon > 0$ and dividing by $n > 0$ gives $14/n < \varepsilon$. Hence

$$\left|\frac{n+6}{n^2-6} - 0\right| \leq \frac{14}{n} < \varepsilon \qquad \text{for every integer } n > N.$$

Conclusion. The threshold supplies the estimate required by Definition 7.1, so

$$\boxed{\frac{n+6}{n^2-6} \longrightarrow 0}.$$

We did not require the first few terms to have a positive denominator.

8.2 Exercise 8.2: Determine five limits and justify each one formally

What we must show. Find the limits of the five stated sequences and prove every answer directly from Definition 7.1. The candidate for a quotient of linear expressions can be suggested by its leading coefficients, but the error calculations below are what establish the limits.

Part (a), Step 1: estimate the difference from zero.

For $a_n = n/(n^2+1)$, the numerator and denominator are positive for every $n \in \mathbb{N}$. Since $n^2 + 1 \geq n^2 > 0$,

$$\begin{aligned} |a_n - 0| &= \frac{n}{n^2+1}, \\ &\leq \frac{n}{n^2}, \\ &= \frac{1}{n}. \end{aligned}$$

We will prove that 0 is the limit.

Part (a), Step 2: supply the quantified proof.

Let $\varepsilon > 0$ and set

$$N = \max\left\{1, \frac{1}{\varepsilon}\right\}.$$

For every integer $n > N$, we have $n > 1/\varepsilon$. Because ε and n are positive, this implies $1/n < \varepsilon$. The error bound from Step 1 then gives

$$|a_n - 0| \leq \frac{1}{n} < \varepsilon.$$

Conclusion. The limit is $\boxed{\lim_{n\to\infty} \frac{n}{n^2+1} = 0}$, as verified for every positive error tolerance.

Part (b), Step 1: compute the exact difference from $7/3$.

Let $b_n = (7n-19)/(3n+7)$. Since $n \geq 1$, its denominator satisfies $3n+7 \geq 10 > 0$. Compute

$$\begin{aligned} b_n - \frac{7}{3} &= \frac{3(7n-19) - 7(3n+7)}{3(3n+7)}, \\ &= \frac{21n - 57 - 21n - 49}{9n+21}, \\ &= \frac{-106}{9n+21}. \end{aligned}$$

Thus

$$\left|b_n - \frac{7}{3}\right| = \frac{106}{9n+21} \leq \frac{106}{9n},$$

because $9n + 21 \geq 9n > 0$.

Part (b), Step 2: choose and verify a threshold.

Given any $\varepsilon > 0$, choose

$$N = \max\left\{1, \frac{106}{9\varepsilon}\right\}.$$

Take any integer $n > N$. Then $n > 106/(9\varepsilon)$, so $9\varepsilon n > 106$ and hence $106/(9n) < \varepsilon$. It follows that

$$\left|b_n - \frac{7}{3}\right| = \frac{106}{9n+21} \leq \frac{106}{9n} < \varepsilon.$$

Conclusion. Since this works for every $\varepsilon > 0$,

$$\boxed{\lim_{n\to\infty} \frac{7n-19}{3n+7} = \frac{7}{3}}.$$

Part (c), Step 1: compute the exact difference from $4/7$.

Let $c_n = (4n+3)/(7n-5)$. Its denominator is positive for all natural indices, since $7n-5 \geq 7-5 = 2 > 0$. Cross-multiplication gives

$$\begin{aligned} c_n - \frac{4}{7} &= \frac{7(4n+3) - 4(7n-5)}{7(7n-5)}, \\ &= \frac{28n + 21 - 28n + 20}{49n - 35}, \end{aligned}$$

$$= \frac{41}{49n - 35}.$$

The error is already positive, so its absolute value has the same expression.

Part (c), Step 2: bound the denominator below by a multiple of n.

For $n \geq 1$,

$$(7n - 5) - 2n = 5n - 5 = 5(n - 1) \geq 0.$$

Consequently $7n - 5 \geq 2n > 0$, and multiplying by $7 > 0$ gives $49n - 35 \geq 14n > 0$. Hence

$$\left| c_n - \frac{4}{7} \right| = \frac{41}{49n - 35} \leq \frac{41}{14n}.$$

Part (c), Step 3: prove the definition with an explicit threshold.

Let $\varepsilon > 0$ and put

$$N = \max\left\{1, \frac{41}{14\varepsilon}\right\}.$$

For an arbitrary integer $n > N$, the inequality $n > 41/(14\varepsilon)$ gives $14\varepsilon n > 41$ and thus $41/(14n) < \varepsilon$. Therefore

$$\left| c_n - \frac{4}{7} \right| \leq \frac{41}{14n} < \varepsilon.$$

The estimates used here hold for every such index, not merely for a selected collection of large indices.

Conclusion. The limit is

$$\boxed{\lim_{n\to\infty} \frac{4n + 3}{7n - 5} = \frac{4}{7}}.$$

Part (d), Step 1: simplify the difference from $2/5$.

For $d_n = (2n + 4)/(5n + 2)$, the denominator satisfies $5n + 2 \geq 7 > 0$. The exact calculation is

$$\begin{aligned} d_n - \frac{2}{5} &= \frac{5(2n + 4) - 2(5n + 2)}{5(5n + 2)}, \\ &= \frac{10n + 20 - 10n - 4}{25n + 10}, \\ &= \frac{16}{25n + 10}. \end{aligned}$$

Thus

$$\left| d_n - \frac{2}{5} \right| = \frac{16}{25n + 10} \leq \frac{16}{25n},$$

where the comparison uses $25n + 10 \geq 25n > 0$.

Part (d), Step 2: choose a threshold and verify the required error.

For an arbitrary $\varepsilon > 0$, set

$$N = \max\left\{1, \frac{16}{25\varepsilon}\right\}.$$

If an integer $n > N$, then $n > 16/(25\varepsilon)$. Multiplication by $25\varepsilon > 0$ gives $25\varepsilon n > 16$; division by $25n > 0$ gives $16/(25n) < \varepsilon$. Hence

$$\left| d_n - \frac{2}{5} \right| \leq \frac{16}{25n} < \varepsilon.$$

Conclusion. Definition 7.1 gives

$$\boxed{\lim_{n\to\infty} \frac{2n+4}{5n+2} = \frac{2}{5}}.$$

Part (e), Step 1: bound the error using the elementary sine bound.

The sequence is $s_n = (\sin n)/n$. For every real x, the elementary trigonometric inequality $-1 \leq \sin x \leq 1$ gives $|\sin x| \leq 1$. Applying it at $x = n$ and using $n > 0$ yields

$$\begin{aligned} |s_n - 0| &= \left|\frac{\sin n}{n}\right|, \\ &= \frac{|\sin n|}{n}, \\ &\leq \frac{1}{n}. \end{aligned}$$

The bound holds for every natural n regardless of the sign of $\sin n$.

Part (e), Step 2: carry out the formal proof.

Let $\varepsilon > 0$ and choose

$$N = \max\left\{1, \frac{1}{\varepsilon}\right\}.$$

For every integer $n > N$, multiplication of $n > 1/\varepsilon$ by $\varepsilon > 0$ and division by $n > 0$ give $1/n < \varepsilon$. Consequently

$$|s_n - 0| \leq \frac{1}{n} < \varepsilon.$$

Conclusion. Thus $\boxed{(\sin n)/n \longrightarrow 0}$. The proof uses a uniform bound on sine, not a claim that the sequence $(\sin n)$ has a limit.

8.3 Exercise 8.3: Taking square roots when the limit is zero

What we must show. Suppose $s_n \geq 0$ for every $n \in \mathbb{N}$ and $s_n \to 0$. Prove that $\sqrt{s_n} \to 0$ directly from Definition 7.1.

Plan. To make the nonnegative square root smaller than ε, it suffices to make the original number smaller than ε^2. The input error tolerance for the given convergence will therefore be $\delta = \varepsilon^2$.

Step 1: choose the input tolerance and use the hypothesis.

Let $\varepsilon > 0$ be arbitrary. Then

$$\delta = \varepsilon^2 > 0.$$

The assertion $s_n \to 0$ means that the definition applies to every positive tolerance, including this particular δ. Hence there is an N such that every integer $n > N$ satisfies

$$|s_n - 0| < \delta = \varepsilon^2.$$

Because $s_n \geq 0$, its absolute value is itself, and the last estimate is exactly

$$0 \leq s_n < \varepsilon^2.$$

Step 2: justify the strict comparison of square roots.

Fix any integer $n > N$ and put $u = \sqrt{s_n}$. The nonnegative square root exists because $s_n \geq 0$, and it satisfies $u \geq 0$ and $u^2 = s_n$. If $u \geq \varepsilon$, then

$$u^2 - \varepsilon^2 = (u - \varepsilon)(u + \varepsilon) \geq 0,$$

since $u - \varepsilon \geq 0$ and $u + \varepsilon > 0$. This would imply $s_n = u^2 \geq \varepsilon^2$, contradicting the estimate from Step 1. Thus $u < \varepsilon$.

Step 3: verify the target absolute-value estimate.

Since $u \geq 0$,

$$|\sqrt{s_n} - 0| = \sqrt{s_n} = u < \varepsilon \qquad \text{for every integer } n > N.$$

The threshold N was supplied by the given convergence for the tolerance ε^2; it does not depend on the later index n.

Conclusion. For every target tolerance $\varepsilon > 0$ we obtained the required threshold, proving $\boxed{\sqrt{s_n} \longrightarrow 0}$. No division by $\sqrt{s_n}$ was used, so the argument includes terms equal to zero.

8.4 Exercise 8.4: A sequence tending to zero times a bounded sequence

What we must show. Assume (t_n) is bounded, so there is a real number M with $|t_n| \leq M$ for every $n \in \mathbb{N}$, and assume $s_n \to 0$. Prove $s_n t_n \to 0$ directly from the definition.

Step 1: choose a positive bound that can be used as a divisor.

The given bound satisfies $M \geq |t_1| \geq 0$, so $M \geq 0$. It need not be strictly positive. Define instead

$$B = \max\{1, M\}.$$

Then $B \geq 1 > 0$, and for every n,

$$|t_n| \leq M \leq B.$$

The constant B is fixed for the whole sequence; it does not depend on the index n or on our later choice of error tolerance.

Step 2: ask the sequence tending to zero for a sufficiently small error.

Let $\varepsilon > 0$ be arbitrary and put

$$\delta = \frac{\varepsilon}{B} > 0.$$

Since $s_n \to 0$, there is an N such that, for every integer $n > N$,

$$|s_n - 0| < \delta = \frac{\varepsilon}{B}.$$

Step 3: verify the product estimate term by term.

Take an arbitrary integer $n > N$. Multiplication of $|t_n| \leq B$ by the nonnegative number $|s_n|$ gives $|s_n|\,|t_n| \leq B\,|s_n|$. Multiplication of $|s_n| < \varepsilon/B$ by the strictly positive number B preserves the strict inequality. Therefore

$$|s_n t_n - 0| = |s_n|\,|t_n|\,,$$

$$\begin{aligned} &\leq B\,|s_n|\,, \\ &< B\frac{\varepsilon}{B}, \\ &= \varepsilon. \end{aligned}$$

This bound holds for every integer $n > N$.

Step 4: check the possible zero-bound case.

If the original M was zero, then $0 \leq |t_n| \leq 0$ forces $t_n = 0$ for every n. The products are then all zero. Our proof already covers this case because it used $B = \max\{1, 0\} = 1$ rather than dividing by $M = 0$.

Conclusion. The error estimate establishes $\boxed{s_n t_n \longrightarrow 0}$ by Definition 7.1. The sequence (t_n) need only be bounded; its own convergence was not assumed or used.

8.5 Exercise 8.5: The squeeze lemma and an absolute-value comparison

What we must show. Prove (a) that $a_n \leq s_n \leq b_n$ for all n, together with $a_n \to s$ and $b_n \to s$, implies $s_n \to s$; and (b) that $|s_n| \leq t_n$ for all n, together with $t_n \to 0$, implies $s_n \to 0$.

Part (a), Step 1: obtain two thresholds for the same tolerance.

Let $\varepsilon > 0$ be arbitrary. Because $a_n \to s$, there is an N_1 such that

$$n > N_1 \quad \Longrightarrow \quad |a_n - s| < \varepsilon.$$

Because $b_n \to s$, there is an N_2 such that

$$n > N_2 \quad \Longrightarrow \quad |b_n - s| < \varepsilon.$$

Each implication is required for every integer index exceeding the corresponding threshold.

Part (a), Step 2: make both estimates valid at once.

Choose

$$N = \max\{N_1, N_2\}.$$

For an arbitrary integer $n > N$, we have both $n > N_1$ and $n > N_2$. Thus both error estimates apply at this same index. The elementary absolute-value equivalence from Exercise 3.7(a) gives

$$-\varepsilon < a_n - s < \varepsilon, \qquad -\varepsilon < b_n - s < \varepsilon.$$

Part (a), Step 3: insert the middle term between these errors.

The order assumption holds for every n, so in particular for our chosen $n > N$ we have $a_n \leq s_n \leq b_n$. Add $-s$ throughout:

$$a_n - s \leq s_n - s \leq b_n - s.$$

Combine the strict lower bound from a_n with the strict upper bound from b_n:

$$-\varepsilon < a_n - s \leq s_n - s \leq b_n - s < \varepsilon.$$

The middle inequalities are permitted to be weak. The outer strict inequalities still give $-\varepsilon < s_n - s < \varepsilon$ by transitivity. Applying Exercise 3.7(a) once more gives $|s_n - s| < \varepsilon$.

Part (a), Step 4: record the quantifiers of the conclusion.

The number $\varepsilon > 0$ was arbitrary. We chose a single N for that tolerance and verified $|s_n - s| < \varepsilon$ for every integer $n > N$. This is exactly the definition of convergence to s.

Conclusion. The squeeze lemma is

$$\boxed{a_n \leq s_n \leq b_n, \quad a_n \to s, \quad b_n \to s \quad \Longrightarrow \quad s_n \to s}.$$

The two enclosing sequences may approach s from either side; only their common limit and the displayed order comparisons were used.

Part (b), Step 1: extract the sign information contained in the hypothesis.

For each n, $|s_n| \geq 0$. The assumed inequality therefore gives

$$0 \leq |s_n| \leq t_n.$$

In particular, $t_n \geq 0$ for every index. This is a consequence of the hypothesis, not an extra assumption.

Part (b), Step 2: apply the definition to the given sequence.

Let $\varepsilon > 0$ be arbitrary. Since $t_n \to 0$, there is an N such that every integer $n > N$ satisfies

$$|t_n - 0| < \varepsilon.$$

As $t_n \geq 0$, this is the same as $t_n < \varepsilon$.

Part (b), Step 3: finish the required error estimate.

For every integer $n > N$, combine Step 1 with Step 2:

$$0 \leq |s_n - 0| = |s_n| \leq t_n < \varepsilon.$$

Thus $|s_n - 0| < \varepsilon$ for every such n.

Conclusion. For every $\varepsilon > 0$ the preceding threshold works, so $\boxed{s_n \longrightarrow 0}$. This direct proof records why the sign of t_n causes no difficulty.

8.6 Exercise 8.6: Absolute values at zero and an alternating counterexample

What we must show. Prove (a) $s_n \to 0$ if and only if $|s_n| \to 0$, and (b) the sequence $s_n = (-1)^n$ has convergent absolute values but is itself divergent.

Throughout these exercises, convergence is used in the precise form of Definition 7.1: $x_n \to L$ means that for every $\varepsilon > 0$ there is a real number N such that

$$n \in \mathbb{N}, \quad n > N \quad \Longrightarrow \quad |x_n - L| < \varepsilon.$$

A threshold N need not be an integer. Whenever an index above a specified threshold is needed, the Archimedean property supplies a positive integer above it.

Part (a), Step 1: compare the two errors exactly.

For every $n \in \mathbb{N}$, the number $|s_n|$ is nonnegative. Therefore

$$||s_n| - 0| = ||s_n|| = |s_n| = |s_n - 0|.$$

The errors in the two proposed zero limits are the same real number, not merely bounded by each other.

Part (a), Step 2: prove the forward implication.

Assume $s_n \to 0$, and let $\varepsilon > 0$ be arbitrary. The definition supplies N such that

$$n > N \quad \Longrightarrow \quad |s_n - 0| < \varepsilon.$$

For each such index, the identity in Step 1 gives

$$||s_n| - 0| = |s_n - 0| < \varepsilon.$$

Thus the same threshold N verifies convergence of $|s_n|$ to 0.

Part (a), Step 3: prove the reverse implication.

Assume $|s_n| \to 0$, and let $\varepsilon > 0$ be arbitrary. Choose N from this convergence so that

$$n > N \quad \Longrightarrow \quad ||s_n| - 0| < \varepsilon.$$

Using Step 1 again yields

$$|s_n - 0| = ||s_n| - 0| < \varepsilon \qquad (n > N).$$

This is the definition of $s_n \to 0$.

Conclusion. Both implications have been proved with the same error and the same threshold:

$$\boxed{s_n \to 0 \quad \Longleftrightarrow \quad |s_n| \to 0.}$$

Part (b), Step 1: prove convergence of the absolute values.

If $s_n = (-1)^n$, then s_n equals either 1 or -1. Hence $|s_n| = 1$ for every n. For any $\varepsilon > 0$, choose $N = 0$. Every positive integer $n > N$ satisfies

$$||s_n| - 1| = |1 - 1| = 0 < \varepsilon.$$

Therefore $|s_n| \to 1$.

Part (b), Step 2: rule out every possible limit of the original sequence.

Suppose, for a contradiction, that $s_n \to L$ for some real L. Apply the definition with the fixed positive tolerance $\varepsilon = 1$. There is a real N such that

$$n > N \quad \Longrightarrow \quad |s_n - L| < 1.$$

Choose an integer $k > \max\{N, 1\}$. Then both $2k$ and $2k + 1$ are positive integers larger than N. Their terms are

$$s_{2k} = 1, \qquad s_{2k+1} = -1.$$

The triangle inequality gives

$$\begin{aligned} 2 &= |1 - (-1)| \\ &= |(1 - L) + (L - (-1))| \end{aligned}$$

$$\begin{aligned}
&\leq |1 - L| + |L - (-1)| \\
&= |s_{2k} - L| + |s_{2k+1} - L| \\
&< 1 + 1 = 2.
\end{aligned}$$

The equality of the absolute values in the fourth line uses $|-x| = |x|$. The resulting strict inequality $2 < 2$ is impossible.

Conclusion. The absolute-value sequence converges to 1, but $((-1)^n)$ has no real limit. Thus existence of a limit for $|s_n|$ alone does not guarantee existence of a limit for s_n.

8.7 Exercise 8.7: Three divergent sequences

What we must show. Show that none of the following sequences converges:

$$\cos\left(\frac{n\pi}{3}\right), \qquad (-1)^n n, \qquad \sin\left(\frac{n\pi}{3}\right).$$

Plan. We will use only the definition of convergence and the triangle inequality. If a sequence converged to L, then for any chosen $\varepsilon > 0$, all sufficiently late terms would be within ε of L. Any two such terms would satisfy

$$|s_p - s_q| = |(s_p - L) + (L - s_q)| \leq |s_p - L| + |s_q - L| < 2\varepsilon.$$

For each proposed sequence, we will produce two arbitrarily late terms whose distance contradicts this inequality. This displayed observation has just been proved directly, so no theorem about subsequences or bounded convergent sequences is needed.

Part (a), Step 1: select a tolerance under an assumed limit.

Suppose

$$s_n = \cos\left(\frac{n\pi}{3}\right) \to L.$$

With $\varepsilon = 1$, choose N such that every positive integer $n > N$ satisfies $|s_n - L| < 1$.

Part (a), Step 2: find late terms equal to 1 and -1.

Choose an integer $k > \max\{N, 1\}$, and put

$$p = 6k, \qquad q = 6k + 3.$$

Both indices exceed N. The period 2π of cosine gives

$$s_p = \cos(2k\pi) = 1, \qquad s_q = \cos(2k\pi + \pi) = -1.$$

Their distance is therefore $|s_p - s_q| = 2$. The two-late-terms estimate gives

$$2 = |s_p - s_q| \leq |s_p - L| + |s_q - L| < 1 + 1 = 2,$$

a contradiction.

Conclusion. The sequence $(\cos(n\pi/3))$ does not converge.

Part (b), Step 1: use the definition with a fixed tolerance.

Suppose $s_n = (-1)^n n \to L$. Choose N from the definition with $\varepsilon = 1$.

Part (b), Step 2: compute the distance between late even and odd terms.

Choose an integer $k > \max\{N, 1\}$. Then $p = 2k$ and $q = 2k + 1$ are larger than N and

$$s_p = 2k, \qquad s_q = -(2k + 1).$$

Their difference is

$$s_p - s_q = 2k + (2k + 1) = 4k + 1 > 2.$$

Since this difference is positive, its absolute value is also $4k + 1$. But convergence would give

$$4k + 1 = |s_p - s_q| \leq |s_p - L| + |s_q - L| < 1 + 1 = 2.$$

This contradicts $4k + 1 > 2$.

Conclusion. The sequence $((-1)^n n)$ does not converge. The proof uses two explicit terms in every sufficiently late tail, rather than assuming a boundedness theorem from a later section.

Part (c), Step 1: choose a tolerance suited to the sine values.

Suppose

$$s_n = \sin\left(\frac{n\pi}{3}\right) \to L.$$

Choose the positive tolerance $\varepsilon = \sqrt{3}/2$. The definition gives N such that $n > N$ implies $|s_n - L| < \sqrt{3}/2$.

Part (c), Step 2: use two positions in the six-term period.

Choose an integer $k > \max\{N, 1\}$ and let

$$p = 6k + 1, \qquad q = 6k + 4.$$

Both are larger than N. By the period 2π of sine,

$$s_p = \sin\left(2k\pi + \frac{\pi}{3}\right) = \frac{\sqrt{3}}{2}, \qquad s_q = \sin\left(2k\pi + \frac{4\pi}{3}\right) = -\frac{\sqrt{3}}{2}.$$

In particular, $|s_p - s_q| = \sqrt{3}$. The convergence estimates would imply

$$\sqrt{3} = |s_p - s_q| \leq |s_p - L| + |s_q - L| < \frac{\sqrt{3}}{2} + \frac{\sqrt{3}}{2} = \sqrt{3}.$$

This is impossible.

Conclusion. The sequence $(\sin(n\pi/3))$ does not converge. Every sequence in the exercise has now been ruled out by the definition of convergence itself.

8.8 Exercise 8.8: Three radical limits with explicit error bounds

What we must show. Prove the three stated limits:

$$\sqrt{n^2 + 1} - n \longrightarrow 0,$$

$$\sqrt{n^2+n}-n \longrightarrow \frac{1}{2},$$
$$\sqrt{4n^2+n}-2n \longrightarrow \frac{1}{4}.$$

The index n is always a positive integer. In particular, division by n is legitimate, and $\sqrt{n^2}=n$ rather than $-n$.

Plan. Multiplication by a conjugate removes the subtraction of large radicals. We will then bound each *error from its claimed limit* by a positive multiple of $1/n$. The estimates below are ordinary algebraic identities and inequalities, not applications of limit laws.

Part (a), Step 1: rationalize the expression and check its sign.

Let $u_n=\sqrt{n^2+1}-n$. Because $n\geq 1$, both n and $\sqrt{n^2+1}$ are positive. Moreover, $\sqrt{n^2+1}>n$: otherwise squaring a supposed inequality $\sqrt{n^2+1}\leq n$ between nonnegative numbers would give $n^2+1\leq n^2$, a contradiction. Thus $u_n>0$. Its conjugate $\sqrt{n^2+1}+n$ is positive, so

$$\begin{aligned} u_n &= \frac{(\sqrt{n^2+1}-n)(\sqrt{n^2+1}+n)}{\sqrt{n^2+1}+n} \\ &= \frac{(n^2+1)-n^2}{\sqrt{n^2+1}+n} \\ &= \frac{1}{\sqrt{n^2+1}+n}. \end{aligned}$$

Part (a), Step 2: obtain an error estimate valid for every index.

The denominator satisfies $\sqrt{n^2+1}+n\geq 2n>0$. Taking reciprocals of positive quantities reverses the comparison, so

$$|u_n-0|=u_n\leq \frac{1}{2n}.$$

Part (a), Step 3: choose and verify a threshold.

Let $\varepsilon>0$ be arbitrary, and choose

$$N=\frac{1}{2\varepsilon}.$$

For every positive integer $n>N$, multiplying by $2\varepsilon>0$ gives $2\varepsilon n>1$. Division by $2n>0$ gives $1/(2n)<\varepsilon$. Hence

$$n>N \quad\Longrightarrow\quad |u_n-0|\leq \frac{1}{2n}<\varepsilon.$$

This is precisely $u_n\to 0$.

Conclusion. $\boxed{\lim_{n\to\infty}(\sqrt{n^2+1}-n)=0}$. An explicit error bound is $1/(2n)$.

Part (b), Step 1: rationalize and factor out the positive index.

Let $v_n=\sqrt{n^2+n}-n$. Multiplying by its positive conjugate gives

$$v_n=\frac{(n^2+n)-n^2}{\sqrt{n^2+n}+n}=\frac{n}{\sqrt{n^2+n}+n}.$$

Set

$$r_n = \sqrt{1 + \frac{1}{n}}.$$

The quantity nr_n is positive, and its square is

$$(nr_n)^2 = n^2\left(1 + \frac{1}{n}\right) = n^2 + n.$$

By uniqueness of the nonnegative square root, $\sqrt{n^2+n} = nr_n$. Substitution gives the exact identity

$$v_n = \frac{n}{nr_n + n} = \frac{1}{r_n + 1}.$$

The number r_n is greater than 1, because $r_n^2 = 1 + 1/n > 1$. Thus $r_n + 1 > 2$, and $0 < v_n < 1/2$.

Part (b), Step 2: rationalize the error from $1/2$.

The sign information in Step 1 gives

$$\begin{aligned}\left|v_n - \tfrac{1}{2}\right| &= \frac{1}{2} - \frac{1}{r_n + 1} \\ &= \frac{(r_n + 1) - 2}{2(r_n + 1)} \\ &= \frac{r_n - 1}{2(r_n + 1)}.\end{aligned}$$

A second conjugate calculation yields

$$r_n - 1 = \frac{(r_n - 1)(r_n + 1)}{r_n + 1} = \frac{r_n^2 - 1}{r_n + 1} = \frac{1/n}{r_n + 1}.$$

Therefore

$$\left|v_n - \tfrac{1}{2}\right| = \frac{1}{2n(r_n + 1)^2}.$$

Since $r_n \geq 1$, we have $(r_n + 1)^2 \geq 4$, so

$$\left|v_n - \tfrac{1}{2}\right| \leq \frac{1}{8n}.$$

Part (b), Step 3: verify the definition of convergence.

Let $\varepsilon > 0$ be arbitrary and choose $N = 1/(8\varepsilon)$. For every positive integer $n > N$, multiplication and division by positive quantities give $1/(8n) < \varepsilon$. Consequently

$$\left|v_n - \tfrac{1}{2}\right| \leq \frac{1}{8n} < \varepsilon \qquad (n > N).$$

The definition proves the desired limit.

Conclusion. $\boxed{\lim_{n\to\infty}(\sqrt{n^2+n} - n) = \tfrac{1}{2}}$. An explicit error bound is $1/(8n)$.

Part (c), Step 1: rationalize and identify a simpler denominator.

Let $w_n = \sqrt{4n^2+n} - 2n$. Its conjugate is positive, and multiplication gives

$$w_n = \frac{(4n^2+n) - 4n^2}{\sqrt{4n^2+n} + 2n} = \frac{n}{\sqrt{4n^2+n} + 2n}.$$

Define $R_n = \sqrt{4 + 1/n}$. Since $n > 0$, the quantity nR_n is the nonnegative square root of $4n^2 + n$:

$$(nR_n)^2 = n^2\left(4 + \frac{1}{n}\right) = 4n^2 + n.$$

Hence

$$w_n = \frac{n}{nR_n + 2n} = \frac{1}{R_n + 2}.$$

Because $R_n^2 = 4 + 1/n > 4$ and $R_n \geq 0$, we have $R_n > 2$. Thus $R_n + 2 > 4$ and $0 < w_n < 1/4$.

Part (c), Step 2: compute and bound the error from $1/4$.

Using the sign of the error, put the fractions over one denominator:

$$\begin{aligned}\left|w_n - \tfrac{1}{4}\right| &= \frac{1}{4} - \frac{1}{R_n + 2} \\ &= \frac{(R_n + 2) - 4}{4(R_n + 2)} \\ &= \frac{R_n - 2}{4(R_n + 2)}.\end{aligned}$$

The second rationalization gives

$$R_n - 2 = \frac{R_n^2 - 4}{R_n + 2} = \frac{1/n}{R_n + 2}.$$

It follows that

$$\left|w_n - \tfrac{1}{4}\right| = \frac{1}{4n(R_n + 2)^2} \leq \frac{1}{64n},$$

because $R_n \geq 2$ implies $(R_n + 2)^2 \geq 16$.

Part (c), Step 3: give the final tolerance-dependent threshold.

Let $\varepsilon > 0$ be arbitrary and choose $N = 1/(64\varepsilon)$. Then $n > N$ implies $1/(64n) < \varepsilon$, so

$$\left|w_n - \tfrac{1}{4}\right| \leq \frac{1}{64n} < \varepsilon \qquad (n \in \mathbb{N},\ n > N).$$

This verifies convergence directly.

Conclusion. $\boxed{\lim_{n\to\infty}(\sqrt{4n^2+n} - 2n) = \tfrac{1}{4}}$. All three limits were proved by explicit inequalities, without using algebraic limit theorems from Section 9.

8.9 Exercise 8.9: Eventual bounds pass to a convergent limit

What we must show. Let $s_n \to s$. Prove:

(a) If $s_n \geq a$ for all but finitely many indices, then $s \geq a$.

(b) If $s_n \le b$ for all but finitely many indices, then $s \le b$.

(c) If $s_n \in [a, b]$ for all but finitely many indices, then $s \in [a, b]$.

A preliminary detail: replace finitely many exceptions by one threshold.

If a property fails only on a finite set $E \subseteq \mathbb{N}$ of indices, define $K = 0$ if E is empty and $K = \max E$ otherwise. Every positive integer $n > K$ is outside E, so every such index satisfies the property. This explains each use of an eventual-bound threshold below.

Part (a), Step 1: suppose the limit violates the eventual lower bound.

Choose K so that $n > K$ implies $s_n \ge a$. Suppose, for a contradiction, that $s < a$. Then the gap $a - s$ is positive. Set

$$\varepsilon = \frac{a - s}{2} > 0.$$

Convergence supplies N_0 such that $n > N_0$ implies $|s_n - s| < \varepsilon$.

Part (a), Step 2: combine the two tail conditions.

Choose a positive integer $n > \max\{K, N_0\}$. The condition $n > K$ gives $s_n \ge a$. The condition $n > N_0$ gives

$$s_n - s < \varepsilon, \qquad s_n < s + \varepsilon = \frac{s + a}{2}.$$

But

$$a - \frac{s + a}{2} = \frac{a - s}{2} > 0.$$

Thus $s_n < (s + a)/2 < a$, contradicting $s_n \ge a$.

Conclusion. The inequality $s < a$ is impossible. Therefore $\boxed{s \ge a}$. Equality $s = a$ is permitted; for example, the constant sequence $s_n = a$ has limit a and satisfies every lower bound in the hypothesis.

Part (b), Step 1: suppose the limit violates the eventual upper bound.

Choose K so that $n > K$ implies $s_n \le b$. Suppose $s > b$, and choose the positive tolerance

$$\varepsilon = \frac{s - b}{2}.$$

Convergence gives N_0 such that $n > N_0$ implies $|s_n - s| < \varepsilon$.

Part (b), Step 2: choose one index after both thresholds.

Take a positive integer $n > \max\{K, N_0\}$. Then $s_n \le b$, whereas the convergence inequality gives

$$-\varepsilon < s_n - s, \qquad s_n > s - \varepsilon = \frac{s + b}{2}.$$

The midpoint exceeds b because

$$\frac{s + b}{2} - b = \frac{s - b}{2} > 0.$$

Hence $s_n > b$, contradicting $s_n \le b$.

Conclusion. The inequality $s > b$ is impossible. Therefore $\boxed{s \le b}$. The endpoint $s = b$ is allowed.

Part (c), Step 1: unpack membership of a closed interval.

For $a \le b$,

$$x \in [a, b] \quad \Longleftrightarrow \quad a \le x \le b.$$

If all but finitely many s_n belong to this interval, then all but finitely many satisfy $s_n \ge a$, and all but finitely many satisfy $s_n \le b$. The finite-exception threshold may be chosen from the indices where interval membership fails, so both inequalities hold beyond the same threshold.

Part (c), Step 2: apply both proved statements to the same limit.

Part (a) gives $s \ge a$, and part (b) gives $s \le b$. Together these say $a \le s \le b$, which is exactly $s \in [a, b]$. If $a = b$, the two inequalities force $s = a = b$, so the singleton case is included as well.

Conclusion. For a convergent sequence,

$$\boxed{s_n \in [a, b] \text{ for all but finitely many } n \quad \Longrightarrow \quad \lim s_n \in [a, b].}$$

The finitely many exceptional early terms do not affect any of these conclusions because an index can be chosen after every exception and after the convergence threshold.

8.10 Exercise 8.10: A strict bound for the limit becomes an eventual strict bound

What we must show. If $s_n \to s$ and $s > a$, prove that there is a real N such that

$$n \in \mathbb{N}, \quad n > N \quad \Longrightarrow \quad s_n > a.$$

Plan. The positive gap $s - a$ allows us to place a whole ε-neighborhood of the limit above a.

Step 1: choose a tolerance smaller than the gap.

Because $s > a$, the real number $s - a$ is positive. Define

$$\varepsilon = \frac{s - a}{2} > 0.$$

The tolerance is fixed from the limit and the bound; it does not depend on the index n.

Step 2: obtain a threshold from convergence.

The definition of $s_n \to s$ gives a real number N such that

$$n \in \mathbb{N}, \quad n > N \quad \Longrightarrow \quad |s_n - s| < \varepsilon.$$

For any index in this tail, the absolute-value inequality implies

$$-\varepsilon < s_n - s < \varepsilon.$$

The left inequality gives $s_n > s - \varepsilon$.

Step 3: check that the lower edge is above the desired bound.

Substitution and simplification give

$$s - \varepsilon = s - \frac{s - a}{2} = \frac{2s - s + a}{2} = \frac{s + a}{2}.$$

Also

$$\frac{s+a}{2} - a = \frac{s-a}{2} > 0,$$

so $(s+a)/2 > a$. Combining these strict comparisons yields

$$s_n > s - \varepsilon = \frac{s+a}{2} > a \qquad (n \in \mathbb{N},\ n > N).$$

Conclusion. The threshold supplied for $\varepsilon = (s-a)/2$ satisfies the required implication:

$$\boxed{n > N \quad \Longrightarrow \quad s_n > a.}$$

The strict hypothesis $s > a$ matters: if $s = a$ and $s_n = a$ for all n, the sequence converges to a but none of its terms is greater than a.